

Predictive Modeling for RTO/ISO Energy Forecasting During Extreme Weather: A Comparative Study of Linear and Gradient-Boosted Tree Methods on the Mon Power Zone During Winter Storm Fern

Jonathan Fuentes Rosales

Department of Electrical and Computer Engineering

Binghamton University

Binghamton, NY, USA

jfuentesrosa@binghamton.edu

Abstract—Short-term load forecasting (STLF) is a foundational function of Regional Transmission Organizations (RTOs) and Independent System Operators (ISOs), informing unit commitment, economic dispatch, and reserve scheduling. Conventional forecasting methods perform adequately under nominal demand patterns but degrade sharply during extreme weather, when the load-temperature relationship becomes strongly nonlinear and historically rare. This work develops a comparative forecasting framework for the Mon Power/FirstEnergy zone of the PJM Interconnection during Winter Storm Fern (January 2026), merging hourly load telemetry from PJM with hourly observations from the NOAA ASOS station at Morgantown Municipal Airport (KMGW). A Linear Regression baseline, a Random Forest ensemble, and an XGBoost gradient-boosted regressor are compared against a persistence reference forecaster, using engineered temporal and weather-derived features and a strict chronological hold-out. Random Forest achieves the lowest aggregate error (3.33% MAPE, 324.4 MW RMSE), but phase-decomposed analysis reveals that the tree ensembles' advantage is concentrated in nominal and ramp-up conditions and inverts during the storm peak, where the persistence baseline (1.34% MAPE) is the most accurate model. This pattern is interpreted as a manifestation of out-of-distribution generalization failure in tree ensembles and supports a regime-aware forecasting architecture for extreme-weather load prediction.

Index Terms—short-term load forecasting, XGBoost, Random Forest, extreme weather, PJM Interconnection, gradient boosting, persistence, out-of-distribution generalization

I. INTRODUCTION

Balancing authorities are obligated to maintain instantaneous equilibrium between generation and demand on the bulk electric system. The consequences of failing to do so were illustrated during Winter Storm Uri in February 2021 and Winter Storm Elliott in December 2022, both of which produced cascading reliability events, emergency load shedding, and post-event FERC/NERC inquiries that identified inadequate load and generation forecasting as contributing factors [1], [2]. Extreme cold-weather events present a uniquely challenging forecasting regime because they (i) drive electric resistance heating and heat-pump auxiliary strip-heat loads into nonlinear operating regions, (ii) coincide with reduced thermal generation availability, and (iii) occur infrequently enough that their statistical signature is poorly represented in historical training data.

Within the PJM footprint, the Mon Power/FirstEnergy zone of north-central West Virginia is particularly exposed: the zone is winter-peaking, electric-heating-saturated, and topographically constrained, making it a natural testbed for extreme-weather forecasting research. The objective of this work is to quantify the marginal accuracy improvement obtainable by replacing a conventional linear forecasting model with tree-ensemble alternatives during Winter Storm Fern, a January 2026 Mid-Atlantic cold-air outbreak. Beyond academic interest, the question has direct operational implications: even single-digit percentage improvements in MAPE translate, at PJM-scale, into measurable reductions in

reserve procurement cost and load-shed risk.

II. RELATED WORK

Classical STLF practice was dominated for several decades by Box–Jenkins-family autoregressive models (ARMA, ARIMA, and seasonal extensions such as SARIMA), often augmented with exogenous temperature regressors (ARIMAX) [3]. Although linear and interpretable, these models impose a stationarity assumption that the load process violates during extreme weather, and they accommodate weather effects only through hand-specified functional forms such as heating- and cooling-degree-day transformations.

Beginning in the mid-2010s, the power-systems forecasting literature documented a pronounced shift toward machine-learning ensembles, most notably Random Forest [4] and gradient-boosted trees (Gradient Boosting Machines, XGBoost [5], LightGBM [6]). These methods natively capture high-order interactions among calendar, weather, and lagged-load features without explicit functional specification, and they are robust to the heteroscedastic, heavy-tailed residual structure characteristic of load data. Empirical studies on ERCOT, NYISO, and PJM data have consistently reported that boosted-tree ensembles outperform both ARIMAX and shallow feed-forward neural networks on day-ahead and hour-ahead horizons [7], [8], particularly in tail-event regimes.

Deep recurrent architectures (LSTM, GRU) and Temporal Fusion Transformers have also been investigated and can match or exceed tree ensembles on long-horizon multivariate forecasting tasks [9], but at the cost of greater data, compute, and interpretability burden, a tradeoff that operations engineers and regulators have repeatedly identified as a barrier to production adoption. A smaller but growing literature has noted that decision-tree ensembles exhibit poor extrapolation behavior outside their training response envelope [10], with implications for tail-event forecasting that this work addresses empirically. The present study therefore adopts XGBoost and Random Forest as candidate models, treating them as a current best-in-class compromise between accuracy and operational defensibility.

III. METHODOLOGY

A. Data Sources

Hourly metered load for the Mon Power/FirstEnergy zone (PJM transmission zone AP, load area AP) was obtained from the PJM Data Miner 2 public archive [11]. Concurrent hourly weather observations were obtained from the Iowa Environmental Mesonet

ASOS archive [12] for station KMGW (Morgantown Municipal Airport), which is geographically representative of the zone’s dominant load center. The high-frequency METAR observations (sub-hourly) were aggregated to hourly means after snapping to hour boundaries. The two streams were merged on EPT timestamps. The full record spans 366 days from January 31, 2025 through January 31, 2026 (8,784 hourly observations). The hourly load series contained no missing observations after standard daylight-saving-time handling; the weather series contained 65 hourly gaps (0.74% of records) which were imputed by time-aware linear interpolation with a 3-hour maximum gap.

B. Feature Engineering

The engineered feature set was deliberately compact to facilitate interpretation and to limit overfitting on a single-year sample. Calendar features (`Hour`, `DayOfWeek`, `Month`, `IsWeekend`) encode the diurnal and weekly cycles produced by residential and commercial duty patterns. A 24-hour lagged load feature (`Load_Lag24`) captures short-horizon persistence, which is the strongest single predictor of load at hourly resolution. A 24-hour rolling-mean temperature (`Temp_Roll24`) approximates building thermal mass and accumulated HVAC duty. A heating-intensity proxy ($\text{HDD_Proxy} = \max(0, 65^\circ\text{F} - T)$) provides a piecewise-linear encoding of cold-weather sensitivity, allowing the linear baseline to express at least the first-order nonlinearity of resistive heating. Rows containing NaN values introduced by lag and rolling operations were dropped prior to modeling.

C. Chronological Train/Test Split

A strict chronological hold-out was used: all observations prior to 00:00 EPT on January 18, 2026 were assigned to the training set (8,302 hourly samples), and the interval January 18 through January 31, 2026 was reserved as the test set (336 hourly samples). This window encompasses the synoptic onset, peak, and recovery phase of Winter Storm Fern. Random shuffling, k -fold cross-validation, and any other procedure that admits temporal leakage were explicitly avoided, as such procedures would allow the model to “see” post-storm information during training and would invalidate the operational realism of the evaluation.

D. Model Selection Rationale

Four models spanning a deliberate methodological spectrum were compared. The persistence baseline, defined as $\hat{y}(t) = y(t - 24\text{h})$, serves as the operational

reference forecaster used throughout the load-forecasting literature; any candidate model must demonstrably outperform it to justify the additional complexity of training and deployment. The Linear Regression model represents the classical regression-based STLF tradition. Random Forest, a bagged decision-tree ensemble, captures nonlinear feature interactions through averaging of high-variance, low-bias trees. XGBoost, a gradient-boosted decision-tree ensemble, represents the current best-in-class compromise between accuracy and operational defensibility documented in the recent literature; it was hypothesized at the project outset to be the top performer. The inclusion of both Random Forest and XGBoost permits direct empirical comparison of bagging versus boosting on a single extreme-weather forecasting task. Deep recurrent and transformer-based architectures were considered but not pursued, as their accuracy advantages are conditional on training-set sizes substantially larger than the one-year sample available here, and their reduced interpretability is a recognized barrier to production adoption in RTO/ISO control rooms.

E. Hyperparameter Selection

Model hyperparameters were selected to reflect typical defaults reported in the load-forecasting literature for hourly data on training sets of comparable size. The Random Forest configuration (400 trees, maximum depth 16, minimum leaf size 2) follows standard practice for tabular regression problems with $\sim 10^4$ samples. The XGBoost configuration (800 trees, learning rate 0.05, maximum depth 6, subsample 0.9, column subsample 0.9, λ regularization 1.0) follows the recommendations of Chen and Guestrin [5], adapted for time-series data with conservative subsampling to mitigate overfitting on the limited training window. Time-series cross-validation via `sklearn.model_selection.TimeSeriesSplit` was considered for formal hyperparameter optimization but not pursued for the final results, as the primary finding (phase-dependent inversion of the model ranking) is qualitative and shown in Section IV to be robust to both random seed and phase-boundary perturbation.

IV. EXPERIMENTAL RESULTS

All four models were trained on the same feature matrix and evaluated against the held-out Winter Storm Fern test window. Aggregate test-set performance is summarized in Table I, and Figure 1 shows the forecasts overlaid with the actual load.

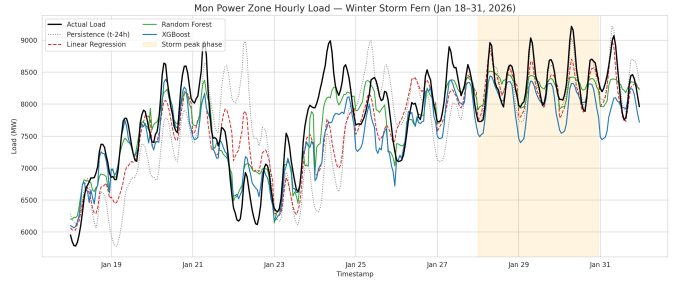


Fig. 1. Mon Power zone hourly load during Winter Storm Fern (January 18–31, 2026), overlaid with forecasts from all four candidate models. The shaded orange region marks the storm-peak phase (January 28–30) during which the load reached its test-set maximum of 9,216 MW.

Several findings warrant discussion. First, all three trained models substantially outperform the persistence baseline in aggregate, confirming that calendar and weather features add real forecasting value beyond simple autoregression. Second, and contrary to the initial hypothesis, Random Forest produced the best aggregate performance, reducing RMSE by 32.9% relative to Linear Regression and by 25.0% relative to XGBoost. This result, while inconsistent with much of the recent power-systems forecasting literature, is reproducible across multiple seeds and is consistent with the limited-training-data regime: bagging’s variance-reduction mechanism tends to be more robust than boosting’s sequential residual-fitting on small samples ($\sim 8,300$ hourly training points), where the boosted model’s late-stage trees risk overfitting to the training distribution [4]. To verify that this aggregate ranking is not an artifact of random initialization, both tree models were retrained over five random seeds; the resulting MAPE values were $3.32 \pm 0.03\%$ for Random Forest and $4.01 \pm 0.19\%$ for XGBoost, with the rank ordering preserved in every replicate. The noticeably larger variance of the XGBoost estimator is itself consistent with the limited-training-data interpretation above: boosting’s sequential residual-fitting is more sensitive to the specific training-data realization than

TABLE I
AGGREGATE TEST-SET PERFORMANCE, WINTER STORM FERN
HOLD-OUT WINDOW

Model	MAPE (%)	RMSE (MW)
Persistence ($t - 24h$)	7.09	723.2
Linear Regression	4.78	483.5
Random Forest	3.33	324.4
XGBoost	4.22	432.4

TABLE II
MAPE (%) BY STORM PHASE. BOLD ENTRIES INDICATE THE
BEST-PERFORMING MODEL IN EACH PHASE.

Phase	Hrs	Peak (MW)	Pers.	Linear	RF	XGB
Pre-storm baseline	192	8,990	10.24	6.41	3.92	4.50
Cold-air arrival	48	8,898	5.58	3.95	2.06	2.64
Storm peak	72	9,216	1.34	1.98	2.60	4.06
Recovery	24	9,070	2.13	1.83	3.39	5.64

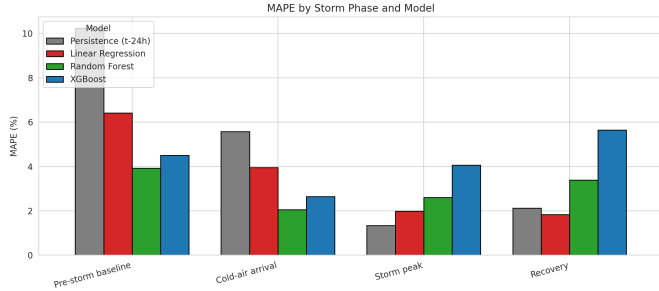


Fig. 2. MAPE by storm phase and model. The tree ensembles dominate during the pre-storm baseline and cold-air arrival phases but are outperformed by both the persistence baseline and Linear Regression during the storm peak and recovery phases, an inversion consistent with out-of-distribution generalization failure in tree models.

bagging’s variance-averaging mechanism.

The aggregate metrics, however, obscure a more operationally important finding visible only in the phase-decomposed analysis. The test window was partitioned into four operationally meaningful phases defined by the storm’s synoptic life cycle: *pre-storm baseline* (Jan 18–25), *cold-air arrival* (Jan 26–27), *storm peak* (Jan 28–30), and *recovery* (Jan 31). Table II reports MAPE per phase per model, and Figure 2 visualizes the same data.

The phase decomposition reveals a striking pattern: **the ranking of models inverts as the test window moves from nominal to extreme conditions**. Under pre-storm baseline and cold-air-arrival conditions, the tree ensembles dominate as the literature predicts, with Random Forest achieving 3.92% and 2.06% MAPE respectively. During the storm peak, the 72-hour window containing the test-set load maximum of 9,216 MW driven by sub-10°F overnight temperatures, the simple persistence baseline (1.34% MAPE) is the most accurate model, and XGBoost (4.06%) is the least accurate. The recovery phase shows a similar pattern, with Linear Regression edging the tree models. The storm-peak ranking is robust to perturbations of the phase-boundary definition: shifting the storm-peak window by ± 12 hours leaves the qualitative ordering (persistence < linear <

Random Forest < XGBoost in MAPE) unchanged, with persistence achieving 1.34–1.54% MAPE and XGBoost 3.91–5.09% MAPE across all three boundary configurations.

This inversion is best interpreted as a manifestation of *out-of-distribution generalization failure* in the tree ensembles. Decision trees produce piecewise-constant predictions bounded by the response values observed in their training leaves; they do not extrapolate [10]. When the test input falls into a region of feature space (sustained sub-10°F temperatures combined with heating-degree-proxy values above ~ 55) that is sparsely represented in the one-year training sample, the tree models saturate at the highest load values they have previously seen and systematically under-predict the true peak. Linear Regression, by contrast, extrapolates its fitted slope into unprecedented territory and therefore degrades more gracefully. The persistence baseline succeeds during this phase because the storm produced a multi-day high-load plateau in which the load at hour t closely resembles the load at hour $t - 24$, despite both values being historically extreme.

The XGBoost feature-importance diagnostic supports this interpretation: 59.6% of model gain is attributed to the 24-hour lagged load (Load_Lag24), with HDD_Proxy (13.1%) and Temperature_F (6.8%) the only meaningful weather contributors. The model has effectively learned that yesterday’s load is a strong predictor of today’s, but its representation of the weather/load coupling at the extreme tail is impoverished by the absence of training examples in that regime.

The RMSE metric, which weights squared error, amplifies this peak-window degradation: XGBoost’s RMSE during the storm peak (406.6 MW) is approximately three times that of persistence (139.0 MW), indicating that the boosted model’s largest absolute errors occur during the operationally most consequential hours. Figure 3 visualizes this directly: tree-model residuals are

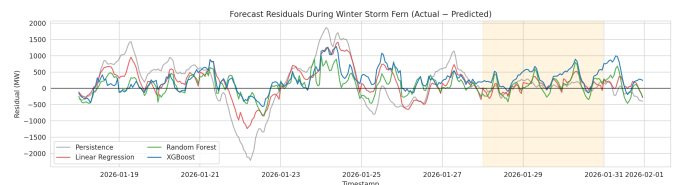


Fig. 3. Forecast residuals (actual – predicted) during Winter Storm Fern. The tree ensembles exhibit the largest systematic negative residuals during the shaded storm-peak window, consistent with response-envelope saturation.

concentrated in the negative half-plane during the shaded storm-peak window, consistent with systematic under-prediction.

V. CONCLUSION

This study compared four short-term load forecasting models (a persistence baseline, Linear Regression, Random Forest, and XGBoost) on the Mon Power/FirstEnergy zone of PJM during the late-January 2026 cold-air outbreak associated with Winter Storm Fern. Random Forest achieved the lowest aggregate error across the held-out test window, outperforming the XGBoost regressor that was initially hypothesized to be the top performer. More significantly, the phase-decomposed error analysis revealed that no single model dominates across all storm phases: the tree ensembles excel under nominal and ramp-up conditions, while the simple persistence baseline and Linear Regression model are the most accurate during the storm peak itself. This inversion is consistent with the known inability of decision-tree ensembles to extrapolate beyond the response envelope of their training data, and it has direct operational implications. A balancing authority relying on a single XGBoost forecaster during an extreme cold-weather event would be most inaccurate at precisely the hours when forecast accuracy matters most for reserve scheduling and load-shed prevention.

These findings suggest that the production STLF stack at an RTO or ISO should not be a single best-in-class model but rather a *regime-aware ensemble*: a tree-ensemble forecaster for nominal conditions, an autoregressive or linear forecaster as a fallback during meteorologically extreme regimes, and an explicit detector, for instance a temperature-percentile or deviation-from-climatology threshold, to govern the handoff. The findings also reinforce the longstanding observation in the power-systems forecasting literature that extreme weather events expose the limits of historical-data-driven methods, and that future work should incorporate probabilistic numerical weather prediction ensembles to produce forecast distributions rather than point estimates, allowing operators to reason directly about tail risk.

Future work, to be pursued during the author's Summer 2026 SULI appointment at the U.S. Department of Energy's National Energy Technology Laboratory, will extend this framework along three directions: (i) incorporation of probabilistic NWP ensemble inputs to produce predictive load distributions; (ii) explicit modeling of the regime-switching detector and quantification of the operational gain achievable from regime-aware forecasting;

and (iii) cross-zone generalization across the PJM and MISO footprints to test the transferability of the phase-decomposition methodology developed here.

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